

Assignment 5

Bilingual Interactive Dashboard



GOALS

This fifth and last assignment is meant to give you hands on experience toward the development of a **small bilingual interactive dashboard** applying concepts seen during *Module 8 – Visualization and Internationalization*.

Your dashboard can be based on any domain/dataset that you wish, either for visualizing food prices, sport statistics, university enrolment, etc.

There are technical requirements for your interactive dashboard site though, so make sure that the domain/dataset you select allows you to meet these requirements.

By carrying out this assignment, you will:

- Reflect on the notion of visualization and the 3Cs (Context, Clutter-free, Contrast) for choosing graph types and designing graphs
- Reflect on internationalization, including translation difficulties and UI elements to localize
- Explore visual and verbal communication in relation to designing a dashboard that will include informative graphs and allow interactivity
- Continue expanding and refining your JavaScript development skills

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SCHEDULE / DEADLINES

Here are the important calendar events for this assignment.

- **Monday July 6th tutorial**

Make sure to attend July 6th tutorial in which you will further explore the requirements for this assignment and the required technologies.

- **Forum monitoring on Brightspace between Tuesday July 7th and Friday July 17th**

If something is not clear, if you need help about the assignment, then please ask questions on Brightspace forum. Make sure you get help. Any question is fine, as long as it is asked in a polite and respectful manner. TAs will be monitoring the forum regularly and answer your questions. Not only will their answers help you, they will help other students in the class.

- **Emails to TAs between Tuesday July 7th and Friday July 17th**

ONLY if the forum help is not sufficient, you can request a one-on-one help session from the TA assigned to your lab session (make sure you contact the TA for YOUR session – Z01 to Z04).

- **Report submission: Sunday, July 19th, midnight**

You MUST submit your report, which will follow the requirements and contain a link to your portfolio by Sunday, July 19th, midnight. *No extension will be given. The only exceptions will be for justified medical reasons with medical certificates. The penalty is 20% per day. There is a grace period till 1am... but then it's -20%!*

- **Group review registration: Sunday, July 19th, midnight**

Make sure you put your name on the group review schedule. You MUST participate in the group review to practice your analysis skills and get feedback on your design.

- **Group review participation between July 20th and 26th**

This time the group review sessions will be scheduled sooner after the submission deadline compared to previous assignments. There are no quizzes to wait for... so we can start sooner. Also, since you have your final exam on July 29th, I want all the group reviews to be done a few days before. Not registering or missing the group review is 20% off your assignment.



ONLINE TUTORIALS / TECHNOLOGIES

Resources for JavaScript development have been provided in the requirement documents for Assignment 1 and 2.

During the July 6th tutorial, ***the TA will provide examples for the development of dashboards.*** I don't see the need for an additional tutorial on July 13th but don't hesitate to ask questions on the forum.



REQUIREMENTS

1. Explore datasets and choose a domain for your analysis

Spend a bit of time exploring datasets. You do not need to have a complex dataset. You can find something online and take a subset of the information for your assignment. The purpose of the assignment is to explore different types of graphs and how to include them in a dashboard. No need for complex data. You can even copy-paste by hand part of some data found online.

If you wish, you can use synthetic data as generated by GenAI software. BUT, if you do that, it is ESSENTIAL that your dashboard indicate clearly that you are presenting synthetic data.

Here are three examples of data that could become source data for an interactive dashboard.

A) Food prices

Statistics Canada publishes information about food prices and shows some visualizations. You can see some charts and info here <https://www.statcan.gc.ca/en/topics-start/food-price> and there is a pointer to the tables ([link in page](#)) which you could use for your own interactive dashboard.

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Geography	Canada (map)		
Products and product groups ^{3,4}	May 2024	April 2025	May 2025
	2002=100		
Vegetables and vegetable preparations	189.7	191.2	193.4
Fresh vegetables	190.5	191.0	192.5
Other fresh vegetables ⁷	223.5	220.9	227.8
Preserved vegetables and vegetable preparations	188.0	192.0	196.8
Frozen and dried vegetables	199.2	198.5	205.2
Canned vegetables and other vegetable preparations	185.1	191.3	195.0

B) Canadian University Enrolment

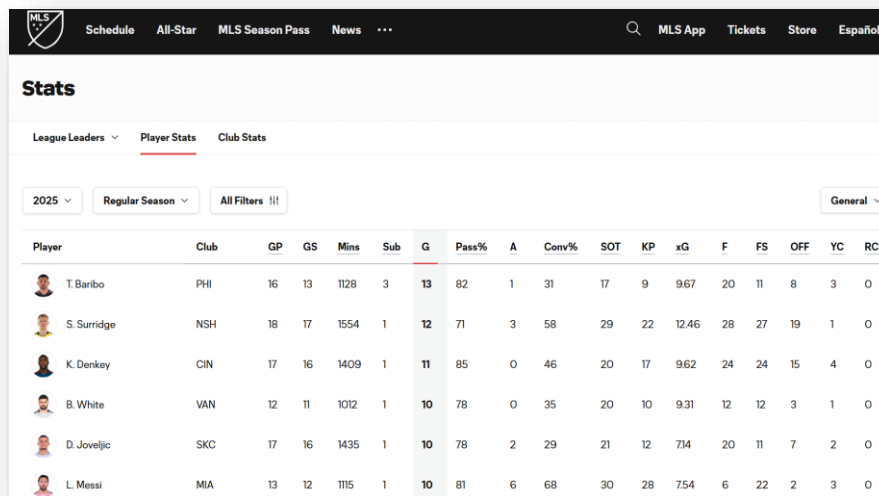
The site Universities Canada has statistics on enrolment per university (<https://univcan.ca/about-universities-canada/facts-and-stats/enrolment-by-university/>). These could be presented in interesting ways in a dashboard, perhaps in complement with some other university related data.

Enrolment by university			
Fall 2024 full-time and part-time fall enrolment at Canadian universities / Sources: Association of Atlantic Universities, Council of Ontario Universities, Individual institutions, Bureau de coopération interuniversitaire			
Search	Ontario		
University ↕	Full-time Undergrad ⇅	Full-time Graduate ⇅	Part-time Undergrad ⇅
▶ Algoma University	5,700	70	480
▶ Brescia University College			
▶ Brock University	15,600	1,600	1,700
▶ Carleton University	19,600	3,900	5,500

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C) Soccer statistics

The MLS publishes some player statistics. <https://www.mlssoccer.com/stats/players/>. I know nothing about sports... but I'm sure all the fan sports among you would find something interesting to display in a dashboard.



The screenshot shows the MLS Stats website interface. At the top, there's a navigation bar with links to Schedule, All-Star, MLS Season Pass, News, and a search icon. Below this, the 'Stats' section is active, with tabs for League Leaders, Player Stats, and Club Stats. The 'Player Stats' tab is selected, and the filters are set to '2025' and 'Regular Season'. A table of player statistics is displayed, with columns for Player, Club, GP, GS, Mins, Sub, G, Pass%, A, Conv%, SOT, KP, xG, F, FS, OFF, YC, and RC. The table lists six players: T. Baribo, S. Surridge, K. Denkey, B. White, D. Joveljic, and L. Messi, with their respective statistics.

Player	Club	GP	GS	Mins	Sub	G	Pass%	A	Conv%	SOT	KP	xG	F	FS	OFF	YC	RC
T. Baribo	PHI	16	13	1128	3	13	82	1	31	17	9	9.67	20	11	8	3	0
S. Surridge	NSH	18	17	1554	1	12	71	3	58	29	22	12.46	28	27	19	1	0
K. Denkey	CIN	17	16	1409	1	11	85	0	46	20	17	9.62	24	24	15	4	0
B. White	VAN	12	11	1012	1	10	78	0	35	20	10	9.31	12	12	3	1	0
D. Joveljic	SKC	17	16	1435	1	10	78	2	29	21	12	7.14	20	11	7	2	0
L. Messi	MIA	13	12	1115	1	10	81	6	68	30	28	7.54	6	22	2	3	0

2. Reflect / Design

Spend time thinking about the various data that would be interesting to show graphically and what graph type would be most appropriate. For this assignment, you are not required to create personas and storyboards. Yet, I do STRONGLY encourage you do mockups even if you will not be required to submit them. In your report, you will be required to describe your design choices, and if you can support those with mockups, that will make your explanations easier.

(A) Decide on 2 charts for visualization

Your interactive dashboard will need to include at least two charts. In the series of videos on Charts, [Choosing Chart Types](#), the presenter (from NN/g) mentions that Bar Chart, Line Chart and Scatter Plots are graph types particularly useful in different situations.

Decide on 2 charts for your dashboard, with a requirement that they be of a different type (e.g. one bar chart and one line chart).

As one example, for the food dataset, the dashboard could present (1) variations in food price for a particular item (e.g. tomatoes) with a line chart and (2) price comparison of different food items (e.g. tomatoes, broccoli, cucumbers) with a bar chart.

(B) Design with the 3Cs

The 3Cs are Context, Clutter-Free and Contrast, as presented in the series of videos on Charts. For each chart, think of its:

- Context (how you will introduce it and what it is meant to show),
- Clutter-Free (what you want to emphasize and what would be *chartjunk*)
- Contrast (how to choose colors, title information and callouts)

(C) Include a title, a proper layout and interactions

The particularity of a dashboard (in contrast to a static report) is that the user should see all the graphs on a single page (or a few pages if there are many graphs) and be able to modify the charts using various menus leading to different selections.

For example, for food price data, the user could choose to display prices in different currencies (e.g. Canadian dollar versus euros), or quantities in different units (e.g. pounds versus kilograms). In the case of sports, the user could vary the temporal extent of a graph and see a player's statistics from 2010-2015 instead of 2020-2025. And in the case of university enrolments, the user could decide which 2 universities to compare. All these possibilities should be offered to the user in the form of menus, toggles or other interactive components.

Also the dashboard should be self-explanatory with a proper title, and a bit of text explaining what the graphs are.

(D) Perform localisation

Your site will need to be in 2 languages. It could be English/French (like Statistics Canada food site example), or you can choose any second language that you want. Only English is mandatory. If you do not speak a second language, no worries, just use Google Translate for your translations. We know it won't be exact, but the purpose in this assignment is not to have perfect translation, but rather for you to witness what will be the impact on your design when it needs to be displayed in 2 languages.

- Make sure to include a Language Selection designed according to the guidelines presented in the localization video.
- For at least ONE chart:
 - Make sure that all textual elements within the chart are in the chosen language (see internationalization video discussing graphs)
 - Make sure that all textual elements outside of the chart (title, interactive elements such as lists and buttons) are in the chosen language.
- Make sure you consider the different elements important in localization (e.g. date, number format, images, icons).

PLEASE NOTE: Although using a resource file is essential for design for internationalization, it will not be mandatory for this assignment provided the timeline and scope of the assignment.

3. Implement your interactive dashboard as a high-fidelity prototype

Using a *JavaScript framework* (e.g., *React*), create a high-fidelity interactive.

The high-fidelity prototype should meet the following technical requirements:

- a. Be developed using a JavaScript framework (e.g., React)
- b. Be interactive (allow a user to actually change some charts with some parameters).
- c. Be accessible through your portfolio

The high-fidelity prototype should be in line with the reflection done in step 2 above:

- (A) The 2 charts should be present and correspond to the chart types decided.
- (B) The 3Cs should be considered
- (C) The layout, title and interactions should be there (at least one interaction per chart)
- (D) The site should be partially bilingual (one chart) with proper language selection

Your high-fidelity prototype should be guided by visual design principles and usability heuristics:

We saw many design guidelines during the semester, try to think of them for your design.

- a. Use a color theme and be consistent
- b. Use typography to attract attention
- c. Think of the impact of the layout and the use of negative space
- d. Use well the principles of contrast, scale, balance, visual hierarchy
- e. Think of the Gestalt Principles for Perception (e.g. similarity, continuity, closure, figure and ground, etc).
- f. Find a good synergy between visual (chart) communication and verbal communication so that both work together.
- g. Think of the 10 usability heuristics and include elements for your prototype to be in line with them.

4. Host your high-fidelity prototype

You are required to host your dashboard online.

5. Link your portfolio to your dashboard and revise your portfolio

You must now link your portfolio to the high-fidelity prototype of the dashboard as you did for all previous assignment.

Your portfolio and dashboard should be accessible to the public (or, if you prefer, password protected with the password provided in your report).

The TA must be able to test your UI without having to communicate with you. This is ESSENTIAL.

6. Revise your portfolio

This is your final assignment, and you are asked to revise and finalize your portfolio to make sure all temporary images that you included in Assignment 1 have been replaced by appropriate images reflecting the choices you made for your service site, memory game, e-commerce site and dashboard.

With the design knowledge you have acquired throughout the semester, your portfolio can certainly look better than it did in Assignment 1. Take the time to have a portfolio that reflects your UI knowledge.

7. Write a short report describing your design process and choices:

Your report should include:

1. Designer: Name + student number
2. Dashboard Goal + Data:
 - a. Provide the domain (sport, food, etc), the dataset (link to source or AI generated) and modifications done to the data (what did you extract or copy/paste from the site).
 - b. Provide links to inspiration dashboards (if any).
3. Reflection/Design:
 - (A) Charts
 - Describe the charts to be shown, and the choice of chart type.
 - (B) 3Cs
 - Describe how you thought of the 3Cs and what you did.
 - (C) Layout, title, interactions
 - Describe the type of interactions (what will the user be able to modify)
 - (D) Internationalization
 - Describe the languages chosen and how you obtained the translations (if you are bilingual or relied on Google Translate)
 - Describe the translation difficulties you encountered (e.g. text size)
 - Describe your localization process (what did you localize?)
4. High-fidelity prototype
 - a. Explain the visual design choices you made.
 - b. Link to your portfolio
5. Code: A link to your GitHub
6. Generative AI Acknowledgement: Many of you will use generative AI to help you create your high-fidelity prototypes and your report. Describe the role of generative AI for each of those, providing the GenAI tools used and your interactions with them.

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EVALUATION

This assignment is worth 10% of your semester and is evaluated on 100 points. The 10% weight reflects the fact that you are both doing the dashboard and revising your portfolio in its final version.

- Your report's content and clarity (5 points)
 - Go through the list of 6 elements given above. Your report should contain all these elements with answers to better understand your design choices and the help provided by GenAI.
 - Make sure the different sections of the report are clearly identified.
 - Avoid long paragraphs, give information in a concise manner, easy to read.
- Your reflection and design choices for the Chart information and types (5 points)
- Your reflection and design choices for the 3Cs (5 points)
- Your reflection and design choices for Internationalization (5 points)
- Your hosted web site, accessible and made according to the requirements provided (50 point)
 - UI should reflect the choices made visual design
 - UI should reflect design choices for charts
 - UI should reflect design choices for internationalization
- Your active participation in the group review (20 points)
 - Ability to analyze other student's design according to the principles explored in the lecture
 - Ability to answer questions from TA and other students
- Your overall portfolio with revised images (10 points)



SUBMISSION CONTENT / PLATFORM

Each student must submit their report, PDF version (see all its requirements above) on Brightspace. The link can be found in the Assignment section, for Assignment 5.

ALSO, very important, **MAKE SURE you register on the signup sheet for your group review.** You will be informed when the signup sheet is available.



QUESTIONS

- See the section Schedule/Deadline of the current document. It indicates tutorials and forum for help and questions.
- Please DO NOT send the professor any emails. The forum help should be enough to obtain answers to your questions. Do not be shy to ask questions on the forum. Any answers provided there by the TA will help many other students.
- If the forum help is not sufficient, email the TA assigned to your lab session (Z01 to Z04)